Inside ONE prompt call, the model should mentally execute this pipeline:

```
1. Read retrieved knowledge
2. Compress it into usable rules
3. Extract observable facts
4. Map facts to rules
5. Apply engineering constraints
6. Reject uncertainty
7. Verify output
8. Return structured JSON
```

STEP 1 — Ingest Retrieved Knowledge

What is provide:

```
[Retrieved Context]
ISA rule...
Tag format...
Symbol meaning...
```

STEP 2 — Compress Knowledge

Long context alone can hurt Context length.

This step is completely dependent on how and how much the data is being retrieved

But we have to make sure that nothing more than necessary is being inputted.

It depends model to model on how much data makes the model overload. So testing is required here

STEP 3 — Extract Observable Facts

(Forces grounding before intelligence)

prompt should say:

```
List ONLY observable inputs.

Do NOT interpret yet.
Do NOT infer.
```

Example output inside the model:

```
Symbol detected: PT
Tag detected: PT-101
Nearby label: Reactor Feed
```

We are separating perception with reasoning

Breaking down tasks to reduce hallucination as the model has more structure to follow

STEP 4 — Rule Mapping (Where reasoning begins)

Now the model connects reality to knowledge.

Prompt instruction:

```
Match each observed fact to a retrieved rule.
Only use retrieved rules.
```

Internal transformation:

```
Observed: PT
Rule: PT = Pressure Transmitter
```

```
Conclusion: Instrument type identified
```

No guessing required.

STEP 5 — Apply Engineering Constraints

Prompt block:

```
RESTRICTIONS:

• Do NOT guess missing fields
• Do NOT invent tag meanings
• Do NOT use knowledge not present in retrieval
• Do NOT infer control loops without evidence
```

What this does inside the model

You shrink its decision space.

Less freedom → fewer hallucinations.

STEP 6 — Uncertainty Handling

Add this instruction:

```
If evidence is insufficient:
Return "REQUIRES_VALIDATION"
```

STEP 7 — Self Verification

Prompt:

```
Before finalizing:

Check:
• Is every field supported by evidence?
• Are rules applied correctly?
• Are there contradictions?
```

We are just making sure that the prompt did not hallucinate or give us wrong information

STEP 8 — Structured Output (Schema Lock)

Prompt:

```
Return ONLY valid JSON.
No explanations.
No extra text.
```

Randomness is the enemy of production AI.

Our model now thinks like this:

```
Load knowledge
↓
Reduce noise
↓
Anchor to facts
↓
Connect facts to rules
↓
Block unsafe guesses
↓
Evaluate certainty
↓
Self-check
↓
Output deterministically
```

Research on Adding examples

1) Examples have shown to improve the accuracy by a considerable amount
2) Adding reasoning on examples help improve eliminate the biases that can be in added examples as the sample size is usually low and it improves accuracy on complex task as the model knows how to think exactly.
3) Long Context alone can hurt accuracy. So We might add just one example in each section and explain the thought process.
4) Modern LLMs do not need COT as they already are reasoning LLMs and it increases the accuracy just by a little bit. However it is still suggested to add reasoning for complex tasks. This further supports the one example approach.